Guide 4 · Demo Playbook

How to run AgriAgent, what to show, what to say, and what to do when it breaks

AgriAgent Final Year Project · 11 October 2026

> **Golden rule.** Always say out loud whether what you are showing is **mock** or **real**. If a real model fails to load, the code silently switches to mock (known issue 5). Check the terminal log before claiming real output.

1 Setup (once)

```
cd finalYearProjectAgriAgent
python3.11 -m venv .venv && source .venv/bin/activate     # Python 3.10 or 3.11
python -m pip install -e '.[dev]'
cp .env.example .env          # put GROQ_API_KEY=... here for real reasoning
```

Checked on 11 October 2026: the app starts and runs in mock mode on the team laptop.

2 Rehearsal 1: mock mode (works today)

```
streamlit run app/app.py      # opens http://localhost:8501
```

1. The sidebar has **Mock Mode** ticked by default. Leave it ticked.
2. Keep "use sample field photo" ticked. The sample is a **synthetic** image.
3. You should see: three boxes, three coloured masks, infestation about 9.4%, savings about 87.8%, and a GeoJSON download with about 10 polygons.
4. Tick **Force System 2**: a prescription card appears (mock diagnosis, Pyrithiobac-sodium for cotton).

What to say: "This is mock mode: fixed stand-in outputs that let us test the pipeline and the safety logic on any laptop. These numbers are not accuracy results."

3 Rehearsal 2: show the safety logic (works today)

This is your strongest live moment, because it shows the rules working.

In mock mode every box takes the **first** phrase of the prompt as its label. Verified results:

Prompt	Labels	Sprayed
`weed . cotton .`	weed ×3	9.4%, 10 polygons
`cotton . weed .`	cotton ×3	0%, no polygons
`cotton weed .`	cotton weed ×3	0%, no polygons

1. Run `weed . cotton .` → weeds are sprayed.
2. Run `cotton . weed .` → labels say "cotton" → nothing is sprayed. The gate never sprays crop.
3. Run `cotton weed .` → still nothing: the label **contains** a crop word, so the gate excludes it. Fail-closed: when in doubt, don't spray.
4. Show the GeoJSON file: only approved weed zones, and say why it is built from masks, not boxes.
5. Optional, in a terminal: `PYTHONPATH=. python -m unittest tests.test_spray_export_safety -v`.

What to say: "Models only propose. These rules, which we test automatically, decide what may be sprayed."

4 Rehearsal 3: real models (to be done after the freeze)

```
python scripts/run_live_pipeline.py --image path/to/field.jpg --prompt "weed . grass ."
```

- First run downloads weights from Hugging Face (needs internet and a few GB of disk).
- **Do not use the script's default prompt**: `"cotton weed . broadleaf plant ."` is rejected by the gate, so nothing is sprayed (known issue 4).
- In the log, confirm you see "Grounding DINO initialized successfully" and **no** "Falling back to mock_mode" line. If you see that line, the output is mock.
- In the app: untick Mock Mode, upload a real photo.

5 The demo script (final version, about 6 minutes)

Min	Show	Say
0–1	Title slide, field photo	Problem: blanket spraying. Goal: spray only weeds.
1–2	Pipeline diagram	Five stages; models advise, rules decide.
2–4	App on a real image (or recorded video)	Boxes → masks → spray map → savings. Point out crop excluded.
4–5	Safety moment (Rehearsal 2)	Change prompt; crop labels never sprayed; tests prove it.
5–6	Results table + the real Kaggle failure	Crop-on-map metric, before vs after our improvement. Honest limits.

6 What can go wrong, and the backup

Problem	Do this
No internet / Groq down	Show mock System 2 and say so; explain the RAG design.
GPU or weights fail	Play the recorded demo video of a real run. Record it the week before.
App crashes	Use `scripts/run_live_pipeline.py` output images saved earlier.
Examiner asks to try their own photo	Allow it in real mode; if results are poor, explain why. That is the research finding.
Laptop dies	Keep the video and slides on a USB stick and in the cloud.

7 Checklist (the day before)

- `git pull` on `main`; all tests pass.
- Real-mode run works offline (weights already downloaded).
- Demo video recorded and copied to two places.
- 3–5 good real images and 1–2 hard ones ready in a folder.
- `.env` with a working Groq key, or a plan to show mock System 2.
- Each team member has rehearsed their part out loud.